GameCapture

GameCapture

Testing Guide for Intercom Dev Team

Capture game audio from iPhone and stream it to Mac
for use as a live audio source in the Intercom app.

Version 1.0 | March 07, 2026
IXG | Confidential

Contents

1.	**Overview**	How GameCapture works
2.	**Architecture**	System diagram and data flow
3.	**Prerequisites**	What you need before testing
4.	**Install the iOS App**	TestFlight setup on iPhone
5.	**Install the Mac Receiver**	Download and run on Mac
6.	**Connect & Stream**	Step-by-step streaming guide
7.	**Using Audio in Intercom**	Route GameCapture audio as a source
8.	**AUDP Wire Protocol**	TCP protocol reference for direct integration
9.	**Recording Audio**	Save captured audio as WAV files
10.	**Troubleshooting**	Common issues and fixes

1. Overview

GameCapture is a two-part system that captures game audio from an iPhone and streams it in real-time to a Mac over a local TCP connection. This allows the Intercom app to use live game audio as a source for broadcasting or mixing.

The system consists of two apps:

- GameCapture (iOS) - Captures all system audio from the iPhone using Apple's ReplayKit Broadcast Extension and streams it over TCP.
- GameCapture Receiver (macOS) - Listens for incoming audio streams, decodes AUDP frames, and plays them through the Mac's audio output.

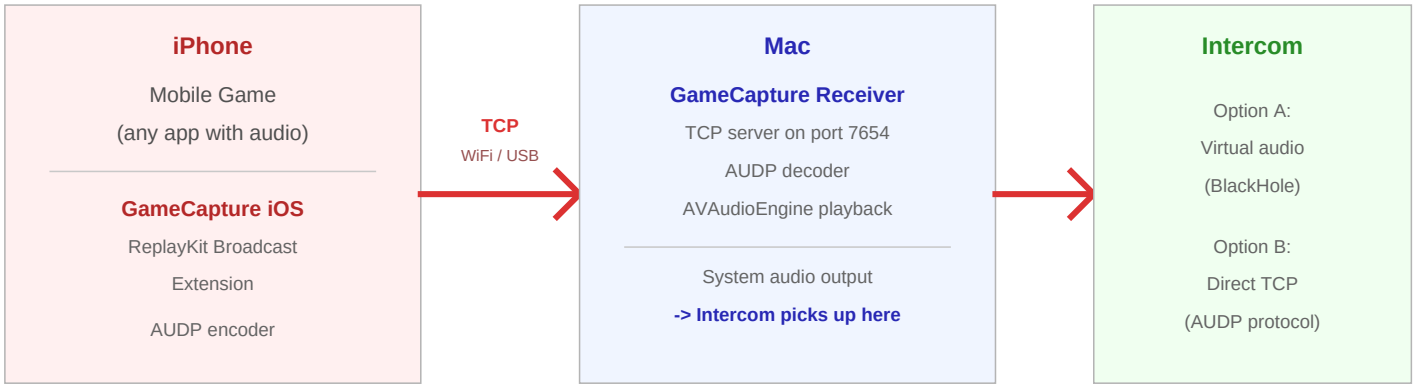
Key Benefits

- Zero latency audio capture from any iOS game
- No jailbreak or special hardware required
- Works over WiFi or USB (USB recommended for lowest latency)
- Uncompressed 16-bit PCM audio at native sample rate
- Built-in WAV recording on the Mac side

2. Architecture

System Diagram

The audio flows through the following pipeline:



Data Flow

- iPhone game produces audio -> ReplayKit captures system audio buffer
- Broadcast extension byte-swaps (big-endian to little-endian on some devices), encodes AUDP frames
- Frames sent over TCP to Mac Receiver's listen address
- Mac Receiver decodes AUDP, feeds PCM into AVAudioEngine for playback
- Intercom app captures the audio via virtual audio device or direct TCP connection

3. Prerequisites

Hardware

- iPhone running iOS 16.0 or later
- Mac running macOS 13.0 (Ventura) or later
- Both devices on the same WiFi network, OR connected via USB cable (recommended)

Software

- TestFlight app installed on the iPhone (free from App Store)
- GameCapture Receiver app on Mac (download from GitHub Releases)

For Virtual Audio Routing (Option A)

- BlackHole (free) - https://existential.audio/blackhole/
- OR Loopback by Rogue Amoeba (paid) - https://rogueamoeba.com/loopback/

A virtual audio driver lets you route GameCapture Receiver's output into the Intercom app as if it were a microphone input. BlackHole is free and works well for this.

For USB Connection (Lowest Latency)

When connected via USB, macOS creates a bridge network interface to the iPhone. The Mac Receiver will show a 'bridge' address in the listen addresses list. Use this address on the iPhone for the lowest latency connection.

> **WARNING**
> USB connection requires the iPhone to be unlocked and trusted on the Mac. You may need to tap 'Trust This Computer' on the iPhone when first connecting.

4. Install the iOS App

The iOS app is distributed via Apple TestFlight.

Step 1: Open this TestFlight invite link on your iPhone:

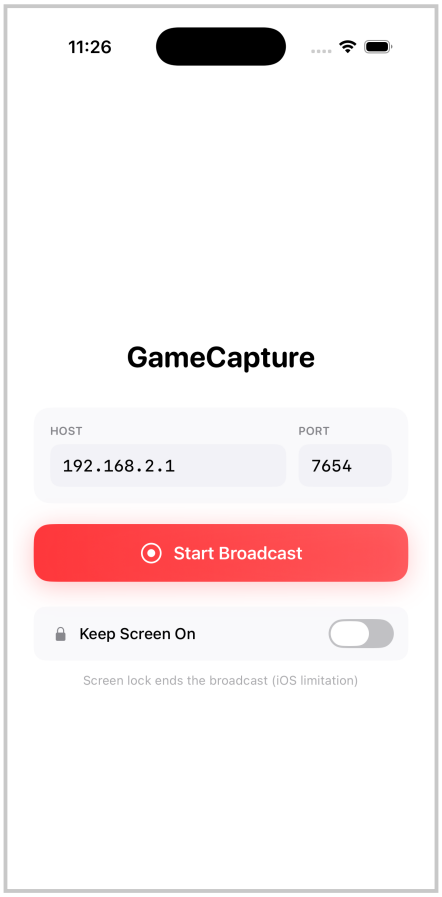
```
https://testflight.apple.com/join/FcGC3Up3
```

Step 2: Tap 'Accept' in TestFlight to join the beta.

Step 3: Tap 'Install' to download the GameCapture app.

Step 4: The app will appear on your home screen with the IXG icon.

iOS App Screenshot



GameCapture iOS app - main screen with host/port configuration

5. Install the Mac Receiver

Download the Mac Receiver from the GitHub Releases page:

```
https://github.com/gowthamchelikam/AudioTap/releases
```

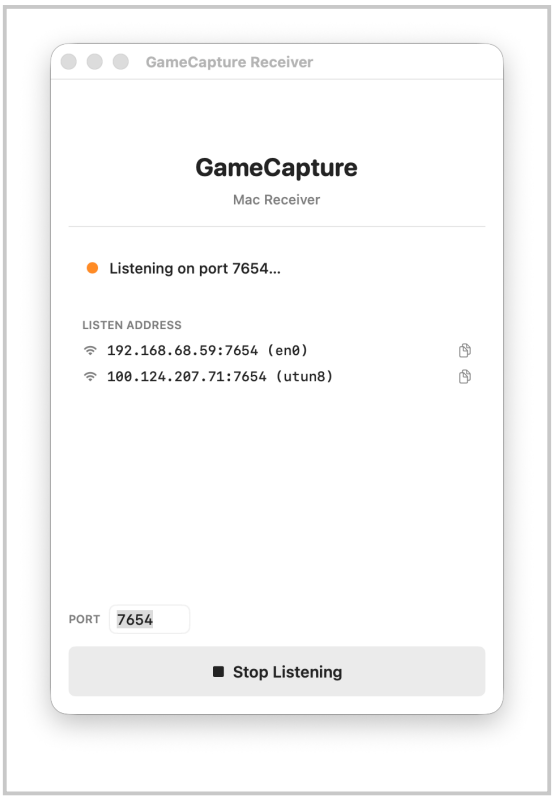
Step 1: Download GameCaptureReceiver-mac.zip from the latest release.

Step 2: Unzip and drag GameCapture Receiver.app to your Applications folder.

Step 3: On first launch, right-click the app and choose 'Open' to bypass Gatekeeper.

Step 4: The app will auto-start listening on port 7654.

Mac Receiver Screenshot



GameCapture Receiver - listening state showing available network addresses

Listen Addresses

The Mac Receiver shows all available network interfaces:
- en0: WiFi address (use this if connecting over WiFi)
- bridge: USB address (use this if iPhone is connected via USB cable)
- utun: Tailscale/VPN address (ignore unless testing remotely)

Click the copy button next to an address to copy the IP to your clipboard.

6. Connect & Stream

Quick Start (5 steps)

Step 1: Launch GameCapture Receiver on your Mac. It auto-starts listening on port 7654. Note the IP address shown (e.g., 192.168.68.59).

Step 2: Open the GameCapture app on your iPhone.

Step 3: Enter the Mac's IP address in the HOST field and ensure PORT is 7654. If using USB, use the bridge address from the Mac app.

Step 4: Tap 'Start Broadcast'. iOS will show the system broadcast picker - select 'AudioTapBroadcast' and tap 'Start Broadcast'.

Step 5: Open any game on the iPhone. Audio will stream to the Mac in real-time. The Mac Receiver will show 'Streaming' status with level meters and format info.

WARNING

IMPORTANT: Do NOT lock your iPhone screen while broadcasting. iOS stops the ReplayKit broadcast when the screen is locked. Enable 'Keep Screen On' toggle in the GameCapture app to prevent this.

Tip: USB Connection

For the lowest latency, connect your iPhone to the Mac via USB cable. The Mac Receiver will display a 'bridge' address (e.g., 192.168.2.1:7654). Enter this address in the iPhone app's HOST field.

7. Using Audio in Intercom

Once GameCapture is streaming, you have two options to use the game audio as a source in the Intercom app:

Option A: Virtual Audio Routing (Recommended)

This is the simplest approach. Use a virtual audio driver to route the Mac Receiver's audio output into a virtual input that the Intercom app can select as a source.

Step 1: Install BlackHole (free virtual audio driver):
https://existential.audio/blackhole/
Choose the 2ch (stereo) version.

Step 2: Open Audio MIDI Setup (Applications > Utilities > Audio MIDI Setup).

Step 3: Click the "+" button at bottom-left and choose "Create Multi-Output Device".

Step 4: Check both "Built-in Output" (or your speakers) and "BlackHole 2ch". This sends audio to both your speakers AND BlackHole simultaneously.

Step 5: Set this Multi-Output Device as your Mac's system output: System Settings > Sound > Output > Multi-Output Device.

Step 6: In the Intercom app, select "BlackHole 2ch" as your audio input source. Game audio from GameCapture will now flow into Intercom.

Option B: Direct TCP Integration

For tighter integration, the Intercom app can directly connect to the GameCapture Receiver's TCP server (or the iPhone's broadcast) and decode the AUDP protocol. See Section 8 for the wire protocol specification.

With direct TCP, the Intercom app acts as its own AUDP receiver, eliminating the need for the Mac Receiver app and virtual audio routing entirely.

Which option to choose?
Option A (Virtual Audio) is fastest to set up for testing. No code changes needed.

Option B (Direct TCP) is better for production. The Intercom app becomes the receiver, giving you full control over the audio pipeline and eliminating the Mac Receiver dependency.

8. AUDP Wire Protocol

If integrating directly via TCP (Option B), here is the complete AUDP wire protocol specification. This is everything you need to build your own receiver.

Connection

The receiver opens a TCP server on a configurable port (default: 7654). The iPhone app connects to this server as a TCP client. Only one connection is supported at a time.

Header (12 bytes, sent once at connection start)

```
Offset  Size     Field        Description
------  ----     -----        -----------
0       4 bytes  magic        0x41554450 ('AUDP') little-endian
4       4 bytes  sampleRate   e.g., 48000 (UInt32, little-endian)
8       2 bytes  channels     e.g., 2 (UInt16, little-endian)
10      2 bytes  bitsPerSamp  e.g., 16 (UInt16, little-endian)
```

Audio Frames (repeated until disconnect)

```
Offset  Size        Field      Description
------  ----        -----      -----------
0       4 bytes     frameSize  Byte count of following PCM data
                               (UInt32, little-endian)
4       frameSize   pcmData    Raw interleaved PCM samples
                               (Int16, little-endian)
```

Pseudocode: Receiving Audio

```
1. Accept TCP connection
2. Read 12 bytes -> parse AUDP header
3. Validate magic == 0x41554450
4. Configure audio output: sampleRate, channels, 16-bit PCM
5. Loop:
   a. Read 4 bytes -> frameSize (UInt32 LE)
   b. Read frameSize bytes -> PCM audio data
   c. Feed PCM to audio engine for playback
6. On disconnect: stop audio, wait for new connection
```

WARNING

BYTE ORDER NOTE: Some iPhone models (e.g., iPhone Air) deliver ReplayKit audio in big-endian Int16 format (CMFormatDescription flags=0xE). The GameCapture iOS app already handles this and always sends little-endian PCM. If you build your own iOS sender, you must detect and byte-swap accordingly.

9. Recording Audio

The Mac Receiver includes a built-in WAV recording feature for capturing game audio to disk.

Step 1: Start streaming game audio (see Section 6).

Step 2: Once connected, a "Record" button appears next to the port field.

Step 3: Click 'Record' to start. A save dialog will appear - choose where to save the WAV file.

Step 4: Click 'Stop Rec' to stop recording. The WAV file is finalized with correct headers.

Recording Format

Files are saved as standard WAV (RIFF) format:
- Sample rate: matches source (typically 48000 Hz)
- Channels: matches source (typically stereo)
- Bit depth: 16-bit signed integer
- No compression (PCM)

These files can be opened in any audio editor (Audacity, Logic, etc.)

10. Troubleshooting

iPhone can't connect to Mac

- Ensure both devices are on the same WiFi network
- Check the Mac's firewall: System Settings > Network > Firewall. Allow incoming connections for GameCapture Receiver
- Try the USB bridge address instead of WiFi
- Verify the IP and port match between iPhone and Mac apps

No audio after connecting

- Make sure a game is running and producing sound on the iPhone
- Check that iPhone volume is not muted (the physical silent switch)
- Check Mac output volume is turned up
- Verify the status shows 'Streaming' (not just 'Connected')

Broadcast stops when screen locks

- This is an iOS limitation. ReplayKit broadcasts stop when the screen locks
- Enable 'Keep Screen On' in the GameCapture iOS app
- Keep the iPhone plugged in to prevent battery drain

Audio sounds like white noise

- This is a known byte-order issue on some iPhone models
- Make sure you're using the latest version of the iOS app (v1.0+)
- The app auto-detects and byte-swaps big-endian audio

Mac Receiver shows 'Listener failed'

- Another app may be using port 7654
- Change the port in the Mac Receiver and update the iPhone app to match
- Or quit any other server that might be using that port

BlackHole not showing as input in Intercom

- Ensure BlackHole is installed correctly (may need a restart)
- Check Audio MIDI Setup shows BlackHole 2ch as a device
- Make sure the Multi-Output Device includes BlackHole
- Set Multi-Output Device as system output in Sound settings

Quick Reference

TestFlight Link	`https://testflight.apple.com/join/FcGC3Up3`
Mac Download	`https://github.com/gowthamchelikam/AudioTap/releases`
Default Port	`7654`
Protocol	`AUDP over TCP (see Section 8)`
Audio Format	`PCM Int16, little-endian, interleaved`
Sample Rate	`Device native (typically 48000 Hz stereo)`
BlackHole	`https://existential.audio/blackhole/`

Questions? Contact the GameCapture team.

IXG | March 2026